

Finite time stabilization of coupled network by the adaptive event-triggered control

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Abstract—This paper studies finite time stabilization of coupled network by an adaptive event-triggered control protocol. Since the resource limitations, continuous control scheme is not valid in practical applications. Event triggered scheme can effectively reduce the cost of control. Unlike existing results, a novel adaptive event-triggered protocol is proposed. It largely reduces transmission times. Finite time stabilization of coupled network is achieved. The theoretical results are confirmed by numerical simulations.

Index Terms—Finite time stabilization; coupled sensing network; adaptive control; event-triggered strategy.

I. INTRODUCTION

Coupled network has received compelling attention due to its abroad applications. Examples include distributed sensor networks [1], [2] and unmanned air vehicles (UAV's) [3]. On the other hand, it is vital for all agents to carry out tasks within a limited time on a coupled network. Most of the methods lead to system asymptotic stability. The convergence speed of such systems can at best be exponential with infinite settling time [4]. In contrast to the asymptotic convergence, finite time convergence requires systems stabilized at equilibrium in a limited time [5], [6], [7], thus performing better application in real life.

In recent decades, many different control schemes have been fully developed to achieve finite time stabilization [8], [9], [10], [11]. A pioneering work of finite time consensus using continuous state feedback is studied in [8]. In [9], the authors investigate sliding mode control to get finite time stabilization. In [10], the authors study finite time consensus based on distributed event-triggered control, where the settling time depends on not only network connectivity and initial condition but also event-triggered condition. Another important control scheme for finite time stability is the adaptive control protocol [11]. Since coupled network usually involves parameter uncertainties, adaptive control scheme for model uncertainties of networks can be tolerated. Hence, it is meaningful to investigate the adaptive control to the finite time stabilization of coupled network.

Control cost is another key factor to design control protocol. Under the event-triggered strategy, the state of system is

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required to sample when some errors exceed the threshold. Hence it can decrease the needs of control actuation. Pioneering works are done in [12], [13]. In [12], the authors investigate the event-triggered control for nonlinear functional systems, where control protocol relaxes more execution requirements. Similar switch system stability, a event-triggered strategy is proposed based on Lyapunov function in [13]. Recently, interesting results as to event-triggered strategy are shown in [14], [15], [16], [17].

Inspired above discussion, this paper explores the finite time stabilization of coupled network by an adaptive event-triggered control. The main contributions are described as follows.

- 1) In comparison with [11], [18], we propose a novel adaptive event-triggered control protocol to achieve finite time stabilization on a directed network. The proposed event-triggered strategy can effectively reduce the communication loads and control cost.
- 2) The proposed control gain lies in a certain set, and it reduces large transients that may occur. Hence, the proposed adaptive event-triggered protocol for network systems is desirable.

We organize the rest of this paper as follows. Some preliminary results about algebraic graph theory and problem formulation of coupled network are contained in Section II. In Section III, sufficient conditions are derived to achieve finite time stabilization. An example illustrating the results is submitted in Section IV. Finally, concluding remarks are presented in Section V.

Notations: $\mathbb{R}$ and $\mathbb{R}^{n \times n}$ represent the set of real numbers and real matrices, respectively. I_n denotes an identity matrix. Let $\mathbf{1} = [1, \dots, 1]^T$; similar notation is used for $\mathbf{0}$. For $X_i \in \mathbb{R}^{n_i \times p}$, $i = 1, \dots, n$, $\text{col}(X_1, \dots, X_n) = [X_1^T, \dots, X_n^T]^T$. Let $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{\frac{1}{p}}$. $\|\cdot\|$ denotes the Euclidean vector norm or the induced matrix 2-norm. Let $\text{sig}^\alpha(u) = \text{sgn}(u)|u|^\alpha$, where $\text{sgn}(\cdot)$ is a standard signum function, i.e.,

$$\text{sgn}(u) = \begin{cases} 1, & u > 0 \\ 0, & u = 0 \\ -1, & u < 0 \end{cases}$$

II. MODEL DESCRIPTION AND PRELIMINARIES

A. Algebraic Graph Theory

A simple directed graph is considered. Let $\bar{G} = (\bar{V}, \bar{E}, \bar{A})$, where $\bar{V} = \{1, 2, \dots, N\}$ represents the set of nodes; $\bar{E} = \{e_{ij} | i, j \in \bar{V}, i \neq j\}$ denotes the set of edges and $\bar{A} = [a_{ij}]$ denotes a weighted matrix; $a_{ij} > 0$ if $(i, j) \in \bar{E}$, otherwise $a_{ij} = 0$. Laplacian matrix $L = [l_{ij}]$ is defined as $l_{ij} = -a_{ij}$, $l_{ii} = -\sum_{j \neq i} l_{ij}$ for $i \neq j$. Denote $\xi = \text{col}(\xi_1, \dots, \xi_N) \in \mathbb{R}^N$ be the normalized left eigenvector of matrix $\bar{A}$ corresponded to zero eigenvalue (ie., $\bar{A}^T \xi = \mathbf{0}$). Denote $\max\{\xi_1, \dots, \xi_N\} = 1$. Let $\xi_{\min} = \min\{\xi_1, \dots, \xi_N\}$. A directed graph $\bar{G}$ is strongly connected if there has a directed path between any pair of distinct nodes k and j in $\bar{G}$ for $k, j = 1, \dots, N$.

B. Problem Formulation

We consider a single isolated node dynamic of the form

$$\dot{s}(t) = Cs(t) + B\tilde{f}(s(t)) \quad (1)$$

where $s(t) = \text{col}(s_1(t), \dots, s_n(t)) \in \mathbb{R}^n$, $C \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times n}$ is a connection weight matrix, and $\tilde{f}(s) = \text{col}(\tilde{f}_1(s), \dots, \tilde{f}_n(s))$ denotes a continuously differentiable function satisfying $\tilde{f}(0) = 0$.

Consider the following dynamical network with N coupled nodes

$$\dot{x}_i(t) = Cx_i(t) + B\tilde{f}(x_i(t)) + c \sum_{j=1}^N a_{ij} \Gamma x_j(t) + u_i(t), \quad i = 1, \dots, N \quad (2)$$

where $x_i(t) \in \mathbb{R}^n$ is the state vector of i -th node, $c > 0$ is the coupled strength of networks, $\tilde{f}(x_i) = \text{col}(\tilde{f}_1(x_i), \dots, \tilde{f}_n(x_i))$, $u_i(t) \in \mathbb{R}^n$ is the control input, $\bar{A} = [a_{ij}]$ denotes the weighted matrix with sum zero (ie. $a_{ii} = -\sum_{j=1, j \neq i}^N a_{ij}$), and Γ is the inner coupling matrix.

Define error state as follows

$$e_i(t) = x_i(t) - s(t), \quad i = 1, \dots, N. \quad (3)$$

Further, from (1) and (2), we have the following error system

$$\dot{e}_i(t) = Ce_i(t) + B\tilde{f}(e_i(t)) + c \sum_{j=1}^N a_{ij} \Gamma e_j(t) + u_i(t), \quad i = 1, \dots, N \quad (4)$$

where $\tilde{f}(e_i) = \tilde{f}(x_i) - \tilde{f}(s)$, $\tilde{f}(e_i) = \text{col}(\tilde{f}_1(e_i), \dots, \tilde{f}_n(e_i))$, and $e_i = \text{col}(e_{i1}, \dots, e_{in})$.

Assumption 1: For any vector $q_1(t), q_2(t) \in \mathbb{R}^n$, the non-linear function $\tilde{f}(\cdot)$ satisfies

$$|\tilde{f}_i(q_1(t)) - \tilde{f}_i(q_2(t))| \leq \sum_{j=1}^n h_{ij} |q_{1j}(t) - q_{2j}(t)|$$

where Lipschitz coefficient $h_{ij} \geq 0$ for $i = 1, \dots, N$.

Definition 1: System (2) is stabilized at objective state $s(t)$ in finite time, if there exist a controller $u_i(t)$ and a scalar $T > 0$ such that

$$\lim_{t \rightarrow T} \|x_i(t) - s(t)\| = 0$$

and

$$\|x_i(t) - s(t)\| = 0 \quad (5)$$

for all $t \geq T$, where T is the settling time depending on initial value.

III. MAIN RESULTS

In this section, we study finite time stabilization via an adaptive event-triggered control scheme.

A. Adaptive Event-Triggered Control Scheme

We define the following measurement errors of coupled node $x_i(t)$ and the target node $s(t)$

$$\begin{aligned} e_{xi}(t) &= x_i(t_{k'}^i) - x_i(t), \\ e_{si}(t) &= s(t_{k'}^i) - s(t) \end{aligned} \quad (6)$$

for any $t \in [t_{k'}^i, t_{k'+1}^i)$, respectively. Then, we have

$$e_i(t) = e_i(t_{k'}^i) - z_i(t) \quad (7)$$

where $z_i(t) = e_{xi}(t) - e_{si}(t)$.

Event-triggered rule is considered as

$$\|z_i(t)\| \leq \gamma_i \|e_i(t)\|, \quad t \in [t_{k'}^i, t_{k'+1}^i) \quad (8)$$

where $\gamma_i \in \mathbb{R}$ and $\gamma_i > 0$, $i = 1, \dots, N$. Let $\gamma = \max(\gamma_1, \dots, \gamma_N)$.

Consider the adaptive control scheme as follows

$$u_i(t) = -k_i(t)e_i(t_{k'}^i) - k_1 \text{sig}^\mu(e_i(t_{k'}^i)) \quad (9)$$

where $k_1 > 0$ and $0 < \mu \leq 1$.

The update law is designed as

$$\begin{aligned} \dot{k}_i(t) &= \text{Proj}(k_i(t)) \\ &= \begin{cases} 0, & \text{if } (1 - \gamma)k_i(t) \geq \bar{k} \\ \xi_i e_i^T(t_{k'}^i) e_i(t_{k'}^i), & \text{if } (1 - \gamma)k_i(t) < \bar{k} \end{cases} \end{aligned} \quad (10)$$

where $k_i(t) \in \mathbb{R}$ with initial condition $4\bar{k}\gamma^2 \leq k_i(0) \leq \frac{\bar{k}}{1-\gamma}$ and $\bar{k} > 0$, and Proj is a projection operator.

B. Finite Time Stabilization Analysis

Theorem 1: Consider system (2), control algorithm (9), and update law (10), under Assumptions 1. Suppose $\bar{A}$ is an asymmetric irreducible matrix. If there exists a constant γ such that $0 < \gamma < \phi$, where $\phi = \min(\lambda^{\frac{1-\mu}{2\mu+2}}, \frac{1}{2})$ and $\lambda = \frac{\xi_{\min}}{Nn}$, then, system (2) can be stabilized at $s(t)$ in finite time. $\square$

Theorem 2: Under the assumptions of Theorem 1, Zeno behavior can be effectively excluded. In others words, $t_{k'+1}^i - t_{k'}^i$, defined by (8) is a positively lower bound. $\square$

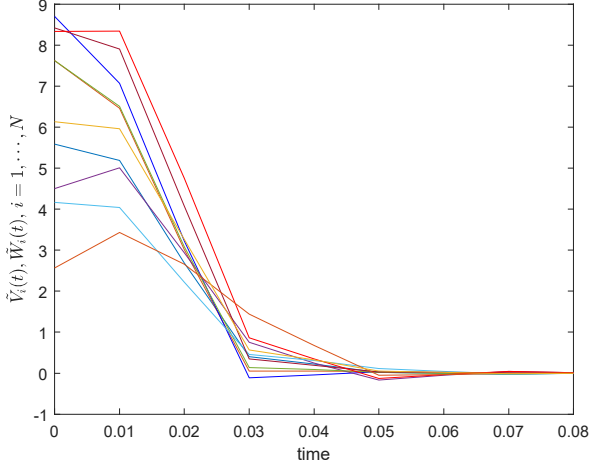


Fig. 1. Profile of $\tilde{V}_i(t)$ and $\tilde{W}_i(t)$ for $i = 1, \dots, N$.

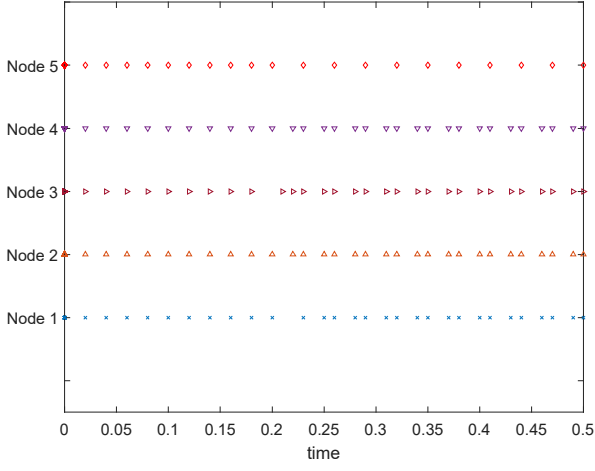


Fig. 2. Profile of triggering times of five nodes.

IV. SIMULATION RESULT

Example 1: An application in the FitzHugh-Nagumo sensing network is considered and a 2-D system [19] is described by

$$\begin{cases} \dot{V} = V - \frac{1}{3}V^3 - W + I_{ex} \\ \dot{W} = \epsilon(V + a - bW) \end{cases} \quad (11)$$

where $V \in \mathbb{R}$ denotes the membrane potential, $W \in \mathbb{R}$ denotes the recovery variable, I_{ex} represents the external stimulus current, and a, b, ϵ are all positive constants. The parameter b has a great effect on the system. We choose parameters $\epsilon = 0.08, a = 0.7, b = 0.8$, and $I_{ex} = 0.5 \cos(t/50)$. The initial conditions $V(0) = -1$ and $W(0) = 0$ are considered. For system (1), we denote $s(t) = \text{col}(V(t), W(t))$, $C = \text{diag}\{0, 0\}$, $B = I_2$, $\tilde{f}_1(V, W) = V - \frac{1}{3}V^3 - W + I_{ex}$, and $\tilde{f}_2(V, W) = \epsilon(V - a - bW)$. By simple computation, system (11) satisfies Assumption 1.

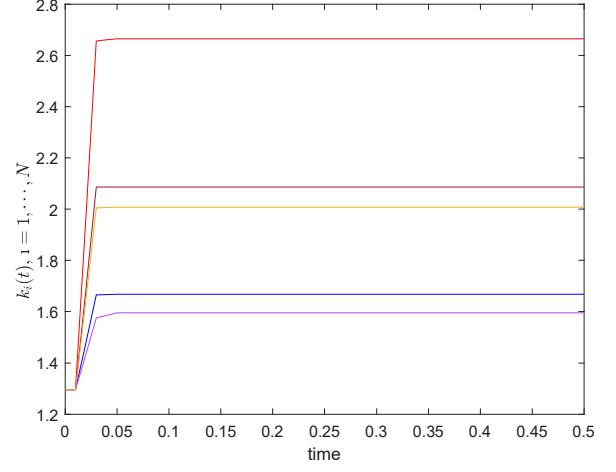


Fig. 3. Profile of adaptive strength.

For system (2), we consider five nodes FitzHugh-Nagumo coupled networks with weighted matrix

$$\bar{\mathcal{A}} = \begin{bmatrix} -1.6 & 0 & 0 & 1.6 & 0 \\ 1.8 & -3 & 0 & 0 & 1.2 \\ 0 & 2 & -2 & 0 & 0 \\ 0 & 0 & 3 & -3 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}.$$

Denote $x_i(t) = (V_i, W_i)$, $C = \text{diag}\{0, 0, 0\}$, $B = I_2$, $\tilde{f}(x_i) = (\tilde{f}_1(V_i, W_i), \tilde{f}_2(V_i, W_i))$. Let $\tilde{V}_i = V_i - V$ and $\tilde{W}_i(t) = W_i - W, i = 1, \dots, 5$. To achieve system (4) stabilized at $s(t)$, the parameters $\gamma = 0.1, k_1 = 1, c = 0.5, \mu = 0.95, k_i(0) = 1.296$, and $\bar{k} = 0.4$ are chosen. The errors of the membrane potential $\tilde{V}_i$ and recovery variable $\tilde{W}_i$ are shown in Fig. 1. Further, we conclude that system (2) is stabilized at $s(t)$ in finite time. Triggering times of control input are plotted in Fig. 2. The adaptive control gain is presented in Fig. 3.

V. CONCLUSION

In this paper, we have explored finite time stabilization of coupled network on a digraph by an adaptive event-triggered control protocol. To realize finite time stabilization, an adaptive event-triggered control protocol was proposed. Simple criterion was provided to guarantee the coupled network in finite time stabilization. Finally, an example was shown the effectiveness of the proposed theoretical results.

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